

Product Description

TPGF-1200 Two-Part Liquid Gap Filler is a thermally conductive, dispensable interface material designed to provide efficient heat transfer between heat-generating components and heat sinks in applications with complex geometries and variable gap thicknesses. The material is engineered for automated or manual dispensing, enabling it to flow and conform to irregular surfaces and achieve low bond line thickness under applied pressure, while minimizing mechanical stress on sensitive components and ensuring consistent thermal contact. TPGF-1200 is a two-component, 1:1 mix ratio silicone-based system formulated with controlled thixotropy and low viscosity to support precise dispensing and gap filling, and cures at room temperature or with heat acceleration to form a stable elastomeric interface for long-term thermal performance.

Key Features

- No pre-mixing or weighing of components required
- Vibration dampening
- Two-part (2k) dispensable material
- Conformable to irregular shapes
- Proven thermal performance with scalable formulations across the TPGF product family
- RoHS Compliant

Applications

- Automotive E-mobility (Converters, Inverters)
- Automotive Infotainment (Display, HUD, Audio)
- Automotive ADAS (DCU, Camera, Radar/Lidar)
- Energy Storage Applications (Residential, Commercial, Grid-Level)
- Power Electronics
- Telecommunications Infrastructure

Application Guidelines

- Avoid exposure to open flame and direct sunlight.
- Ensure application surfaces are clean.
- Avoid air entrapment during dispensing

Storage and Shelf Life

- Store at 8°C to 28°C in a dry environment (RH<70%).
- Shelf life is 12 months from the date of manufacture.
- Compatible with automated and manual dispensing systems

Packaging Information

- Available in 50 cc (150 g) and 400 cc (1 kg) dual cartridge packaging
- Other sizes are available upon request

Property	Value	Test Method
Color (A)	Gray	Visual
Color (B)	Gray	Visual
Mixing Ratio	1:1	--
Thermal Conductivity (W/m·K)	12.0	ASTM D5470
Minimum BLT (µm)	250	--
Hardness (Shore 00)	60	ASTM D2240
Density (g/cc)	3.5	ASTM D792
Viscosity (cps)	350,000	ASTM D374
Operating Temperature Range (°C)	-50 to 200	--
Flammability Rating	V-0	UL94
Volume Resistivity (Ω-cm)	>10 ¹³	ASTM D257
Curing Time @ 25°C (hours)	24	--
Curing Time @ 100°C (minutes)	10	--

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.